SREday Dallas 2026

What Happens at 300 RPS?

Testing an Infrastructure Change Before It Ships

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```
-  default = "t3.micro"
+  default = "t3.large"
```

evolv consulting HQ, Dallas · Fri Oct 23, 2026 · 10:00 AM CDT

A PR review sees the diff, not the behavior

A reviewer can check syntax, naming, and `terraform plan` output, but usually has no evidence of latency, CPU, errors, or cost under traffic.

PR

A typical review thread

“LGTM”

Today's goal: attach a stated, reproducible scenario to the PR, so the review is about evidence, not opinion.

What the diff can't answer

? **Cost**

What does it cost?

? **Capacity**

Will this be enough?

? **Performance**

Latency, CPU, and errors under traffic?

? **Reliability**

What happens if a node dies?

The change we'll test

canvascloudai/le-tf-infra-aws#6

Pritunl VPN server instance_type t3.micro → t3.large

```
variable "instance_type"
- default = "t3.micro"
+ default = "t3.large"
```

Our public fork of binbash's Leverage reference architecture for AWS. Head 3569c3c.

Same one-line change as #5; #6 is the newer report version.

01

tools-vpn-server/variables.tf

Sets the instance_type default: the one touched input. No EC2 resource is named here.

02

A Git module chain carries it

The variable flows through a Git module chain into the VPN server's EC2 module.

03

The EC2 module creates the instance

t3.micro → t3.large on a Pritunl VPN server, not an app tier behind an ALB.

Scenario card · le-tf-infra-aws#6

Define the scenario before you trust a number

Key question for this PR: is 300 RPS of “typical” request traffic meaningful for a Pritunl VPN server? Probably not.

01 Traffic

300 RPS, constant, synthetic. Same on base and head 3569c3c.

02 Duration / steps

Steps 0–29, as shown on the report's per-step table.

300 RPS

offered client traffic

03 App weight

“Typical”, defaulted. Other fields may be assumed.

04 Not evaluated

Live infra, application code, production traffic.

LIVE DEMO: the 300 RPS report on le-tf-infra-aws#6

Workload: synthetic 300 RPS · head 3569c3c

Merge review summary

“No unfavorable changes detected in comparable modeled metrics; modeled cost up $0.14/hr (trade-off).”

01

Check: neutral

Neutral on purpose. See what it doesn't tell you.

02

Steps 0–29

Before: 257 RPS ok, 14.29% errors

After: 300 RPS ok, 0% errors

What this report doesn't tell you

Real performance. Every number is modeled, not measured on a real VPN box.

Your real inputs. Region and resource count were filled in by default; they're listed in the collapsed inputs section.

le-tf-infra-aws#6 · Before → After

Reading the report one dimension at a time

Read the label before the number.

Dimension	Metric	Before (base)	After (PR/head)	Label
Cost	Cost per hour	$0.43/hr	**$0.57/hr (+$0.14)**	modeled
Capacity	Peak aggregate CPU (max of steps)	100%	**41.2%**	modeled
Performance	p99 latency	4073 ms	**381 ms**	modeled
Performance	Goodput	257 RPS	**300 RPS**	modeled
Performance	Error rate	14.29%	**0.00%**	modeled
Reliability	Resilience	0/100	**76/100**	modeled

All values are modeled for a synthetic 300 RPS scenario, not measured on AWS. Source: le-tf-infra-aws#6, report comment 6175470869.

cwm-bench · typical-v1 · 2026-09-27

Should you believe it? How the model is checked

24.3
before fit

→

300 RPS holdout score

82.4 without cost. Later day: 76.0. Second region: 81.1. Three single runs, not an accuracy range.

84.2
after fit

01 Owned benchmark

Real AWS, k6 in-VPC through an internal ALB. Predictions frozen before measurement.

02 Fit vs holdout

Fit at 20 / 100 / 200 RPS. Independent holdout at 300 RPS. 500 is diagnostic, not validated.

03 After fit at 300 RPS

App CPU 14.74% predicted vs 15.30% measured. Goodput 262.60 vs 262.61 RPS.

typical-p95-v1 · preregistered · Oct 10–12

P95, the tail, is now checked too.

Tested at 1–3 servers, 100–300 RPS.

9/9 cells within tolerance

87.6 median P95-only score

Server-count change verdict: passed

1× m5.large

100 RPS **pass** 89.0

200 RPS **pass** 79.3

300 RPS **pass** 87.6

Measured 94.7 ms, model 106.5 ms

2× m5.large

100 RPS **pass** 87.1

200 RPS **pass** 95.7

300 RPS **pass** 89.7

3× m5.large

100 RPS **pass** 99.1

200 RPS **pass** 71.2

300 RPS **pass** 68.8

Lowest scores at 3 servers (still passing)

Pass: measured P95 landed within the tolerance band set before the test.

Next to validate: DB saturation and traffic above 300 RPS.

Replace the stated test

When 300 RPS isn't your workload

Every infra PR: 300 RPS check

Your numbers: API or MCP

Check the label: diagnostic outside 20 to 300 RPS

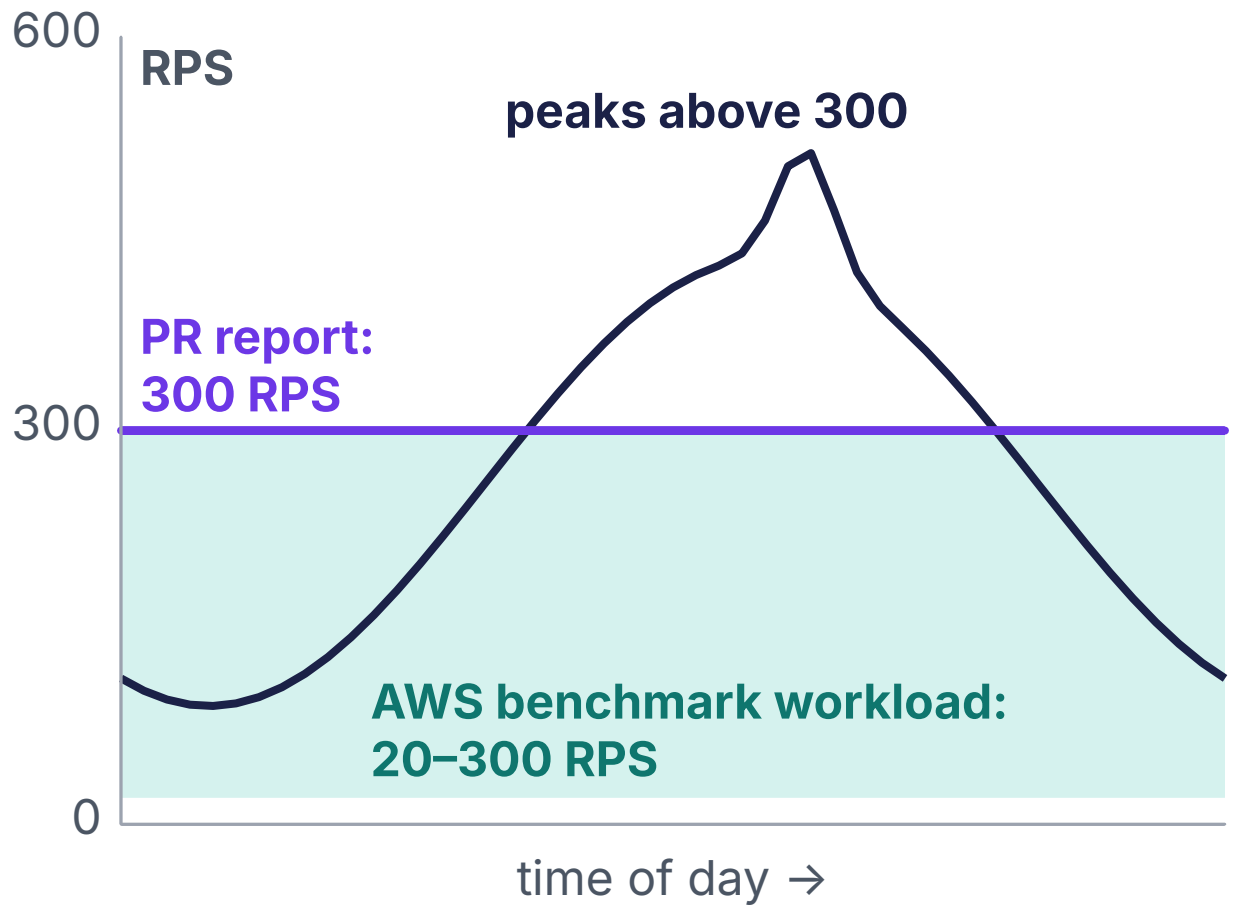
Workload type first

Ask: one real infra PR, blunt feedback.

cloudworldmodel.ai/accuracy

Illustrative, not real data

Your traffic over one day vs the stated test



Take it home

Takeaway: questions to ask in an infra PR review

1 **What scenario was this tested at?**

RPS, duration, traffic shape. Inside the AWS-checked range?

2 **Which numbers are observed, exact, inferred, or assumed?**

Open the report's inputs section: region, size, count.

3 **Which metric does the decision depend on?**

Cost, CPU, P95/P99, errors: how accurate is the model on that one?

4 **What would change the conclusion?**

Rerun at your real peak with the Cloud World Model API or MCP.

Try it on your next infra PR

Setup guide:

cloudworldmodel.ai /docs/github-pr-reports

Thank you · Q&A

Simulates an infra change before merge: modeled cost, p99 latency, peak CPU and resilience, posted on the PR.

Looking for 5 teams: one real infra PR, blunt feedback. No contract.

Install the GitHub App for a 300 RPS check on every infra PR. Use the API or MCP to test at your own traffic.

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cloudworldmodel.ai/teams

github.com/apps/cloud-world-model